PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	The Effect of Aspirin in Takotsubo Syndrome: Protocol of A Systematic Review and Meta-Analysis
AUTHORS	Lin, Jinhai; Wu, Bingxin; Lin, Luoqi; Ding, Yining; Zhong, Biying; Huang, Zhiwei; Lin, Miaoyang; Xu, Dan-Ping

VERSION 1 – REVIEW

REVIEWER	Liu, Tong Tianjin Medical University
REVIEW RETURNED	20-Jan-2021

GENERAL COMMENTS	According to the protocol, this meta-analysis was designed to focus on the long-term effect of aspirin in Takotsubo Syndrome, which is a substantial topic and harbors an potential benefit in clinical setting if doing well. The present protocol described the rationale and procedure of the ongoing meta-analysis clearly. However, there are several vague points which needs to be further elucidate. 1)Patients: according to which kind of TTS diagnostic criteria , patients were thought to be eligible? Or there were no rigorous restrictions for diagnosis of TTS? Whatever it designed to be, I think it would be better to delineate it clearly. 2)Intervention: Since the aim of this meta-analysis was to evaluate the impact of long-term maintenance treatment of aspirin on TTS. So about the treatment duration, how long will be defined as an eligible 'long-term' duration? More definitions about the duration and dosage of aspirin are needed to reduce the potential heterogeneity. 3)Comparison: No detailed interpretation about the comparison was made in the present protocol. No matter observational or interventional study, a clarification of control arm will be better.

REVIEWER	Kato, Ken Chiba University Graduate School of Medicine, Department of Cardiovascular Medicine
REVIEW RETURNED	25-Feb-2021

GENERAL COMMENTS	This is a manuscript of a protocol investigating the effect of aspirin in takotsubo syndrome (TTS). The study design is a systematic review and Meta-analysis. There are several issues which the authors should resolve adequately as listed below.

	1. In the Introduction, the authors mentioned about increased risk of LV thrombus in TTS. However, aspirin is not used for LV thrombus. To solve this clinical question, the effect of anticoagulants in TTS should be investigated. 2. In Search strategy, the authors will search words “Takotsubo Cardiomyopathy” only. However, in the field of TTS research, a number of different names has been used such as takotsubo syndrome, takotsubo cardiomyopathy, tako-tsubo syndrome/cardiomyopathy, ampulla cardiomyopathy, stress cardiomyopathy, apical ballooning syndrome, broken heart syndrome, etc. The search strategy should be reconsidered. 3. In Types of studies, the authors stated that “No limitations are set on the timing of taking aspirin or the length of follow-up.” The aim of this study is to evaluate the long-term effect of aspirin for patients with TTS. To achieve this goal, the timing and duration of taking aspirin are very important. 4. In Exclusion, the authors stated that “Studies including participants with malignancies or autoimmune diseases or taking antithrombotic therapy for other diseases will be excluded.” As a mean age of patients with TTS is around 60-70 years old, a substantial portion of patients should have such conditions. Thus, according to this study design, very few patients with TTS will be included in this Meta-analysis.

VERSION 1 – AUTHOR RESPONSE

Responses to the Reviewers' comments:
Reviewer: 1
1)Patients: according to which kind of TTS diagnostic criteria, patients were thought to be eligible? Or there were no rigorous restrictions for diagnosis of TTS? Whatever it designed to be, I think it would be better to delineate it clearly.
Response: We apologize for the confusion in the diagnostic criteria. As they differ greatly among researches, no specific restriction is set in the criteria of diagnosis. Instead, each adapted criterion and the corresponding bias will be documented and reported in the analysis.

2)Intervention: Since the aim of this meta-analysis was to evaluate the impact of long-term maintenance treatment of aspirin on TTS. So about the treatment duration, how long will be defined as an eligible ‘long-term’ duration? More definitions about the duration and dosage of aspirin are needed to reduce the potential heterogeneity.
Response: Thank you very much for reminding the omission of the duration. Given a lack of comprehensive understanding on the nature of TTS, currently it seems challenging to definite the duration. As TTS is widely considered a special kind of heart failure, we set it to more than 90 days in revision. Also, each dose of Aspirin will be included and analyzed.

3)Comparison: No detailed interpretation about the comparison was made in the present protocol. No matter observational or interventional study, a clarification of control arm will be better.
Response: Thank you for the reminder. We have revised accordingly by adding the comparison between TTS patients taking and not taking Aspirin.

Reviewer: 2
1.In the Introduction, the authors mentioned about increased risk of LV thrombus in TTS. However, aspirin is not used for LV thrombus. To solve this clinical question, the effect of anticoagulants in TTS should be investigated.

Response: Thank you very much for pointing it out. Indeed, anticoagulants are almost irreplaceable for LV thrombus. However, evidence is still lacking under this circumstance. Also, in existing case reports, mostly LV thrombus appears and is treated in 3 months after diagnosis, while the study aims to explore the long-term effect of drug administration, which is set to more than 90 days in this revision. Therefore, we believe that in long term antiplatelet therapy may be favorable and beneficial in practice.

2.In Search strategy, the authors will search words “Takotsubo Cardiomyopathy” only. However, in the field of TTS research, a number of different names has been used such as takotsubo syndrome, takotsubo cardiomyopathy, tako-tsubo syndrome/ cardiomyopathy, ampulla cardiomyopathy, stress cardiomyopathy, apical ballooning syndrome, broken heart syndrome, etc. The search strategy should be reconsidered.
Response: We apologize for the lack of clarity. Subject words and free words of each term will be employed to collectively summarize the findings. For instance, in PubMed, the MeSh term and corresponding entry terms of ‘Takotsubo Syndrome’ will be adapted.

3.In Types of studies, the authors stated that “No limitations are set on the timing of taking aspirin or the length of follow-up.” The aim of this study is to evaluate the long-term effect of aspirin for patients with TTS. To achieve this goal, the timing and duration of taking aspirin are very important.
Response: Thank you very much for pointing it out. Given a lack of comprehensive understanding on the nature of TTS, currently it seems challenging to definite the duration. As TTS is widely considered a special kind of heart failure, the duration is set to more than 90 days in revision. Also, each dose of Aspirin will be included and analyzed.

4.In Exclusion, the authors stated that “Studies including participants with malignancies or autoimmune diseases or taking antithrombotic therapy for other diseases will be excluded.” As a mean age of patients with TTS is around 60-70 years old, a substantial portion of patients should have such conditions. Thus, according to this study design, very few patients with TTS will be included in this Meta-analysis.
Response: Thanks for pointing it out. In fact, the statement serves as an attempt to limit the bias from accompanying diseases that impact the result yet difficult to eliminate. As a result, however, limited individuals may be included in the final analysis. Hence, we will cautiously illustrate the potent effect of our finding.

VERSION 2 – REVIEW

REVIEWER	Liu, Tong Tianjin Medical University
REVIEW RETURNED	26-Apr-2021

GENERAL COMMENTS	No further comments

REVIEWER	Kato, Ken Chiba University Graduate School of Medicine, Department of Cardiovascular Medicine
REVIEW RETURNED	06-May-2021

GENERAL COMMENTS	1.In the Introduction, the authors mentioned about increased risk of LV thrombus in TTS. However, aspirin is not used for LV thrombus. To solve this clinical question, the effect of anticoagulants in TTS should be investigated.

	Response: Thank you very much for pointing it out. Indeed, anticoagulants are almost irreplaceable for LV thrombus. However, evidence is still lacking under this circumstance. Also, in existing case reports, mostly LV thrombus appears and is treated in 3 months after diagnosis, while the study aims to explore the long-term effect of drug administration, which is set to more than 90 days in this revision. Therefore, we believe that in long term antiplatelet therapy may be favorable and beneficial in practice. -> The authors might be misunderstanding my comment. In general, antiplatelet therapy is not applied for treatment of LV thrombus. Therefore, I recommend that the effect of anticoagulants (including warfarin or DOAC), not aspirin, should be investigated, if the authors seek to solve the issue about LV thrombus in TTS.

VERSION 2 – AUTHOR RESPONSE

Responses to the Reviewers' comments:

Reviewer: 1

Reviewer: 2

1. The authors might be misunderstanding my comment. In general, antiplatelet therapy is not applied for treatment of LV thrombus. Therefore, I recommend that the effect of anticoagulants (including warfarin or DOAC), not aspirin, should be investigated, if the authors seek to solve the issue about LV thrombus in TTS.

Response: Please accept our apology for not expressing our idea with clarity. It's true that antiplatelet therapy is not recommended for LV thrombus. However, it is unclear when it comes to TTS without detected thrombus. Also, in clinical practice, many TTS patients are taking Aspirin alone even though its benefit remains uncertain. Therefore, we seek to judge its role in TTS treatment. Undoubtedly, anticoagulants present more potent in this field than antiplatelet drugs and we will further explore their usage to the prognosis of TTS patients if conditions allow. Thank you again for pointing it out.

VERSION 3 – REVIEW

REVIEWER	Kato, Ken Chiba University Graduate School of Medicine, Department of Cardiovascular Medicine
REVIEW RETURNED	29-Jun-2021

GENERAL COMMENTS	None